Motivation: Why Computational-Delay Attacks?

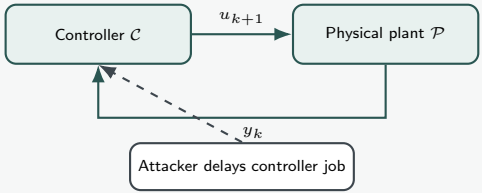
Attack scenario

- A real-time controller must finish each control computation before its deadline.
- An attacker can cause **CPU starvation or execution delays**, making a control job miss its deadline.
- The attacker need not alter the sensor measurement or plant equations. The physical plant may be harmed because a fresh control command is unavailable on time.

Research question

How can an attacker maximize physical disruption while avoiding timing-based detection?

Closed-loop attack path



The plant continues evolving even when the controller misses a deadline.

Separate the contributions

- **Paper:** attack model, optimization formulation, and statistical detector analysis.
- **This project:** physical/interactive demonstration and analysis of defender parameters.

Source: T. Nauta, H. Sandberg, and M. Maggio, "Stealthy Computational Delay Attacks on Control Systems," ICCPS 2025.

Discrete-Time Plant and Controller

Physical plant — Eq. (1)

$$\begin{aligned} x_{k+1} &= Ax_k + Bu_k, \\ y_k &= Cx_k + Du_k. \end{aligned}$$

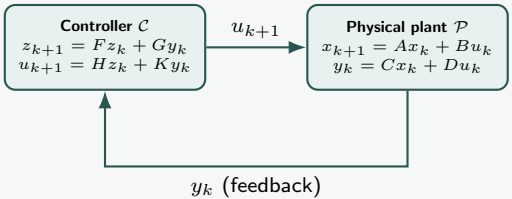
- x_k: plant state
- u_k: applied plant input
- y_k: plant output/measurement

Controller — Eq. (2)

$$\begin{aligned} z_{k+1} &= Fz_k + Gy_k, \\ u_{k+1} &= Hz_k + Ky_k. \end{aligned}$$

- z_k: controller state
- y_k: measured plant output
- u_{k+1}: command for the next period

Complete physical feedback loop



Timing convention

During period k, the plant holds u_k constant. The controller measures y_k and computes u_{k+1} for period $k+1$. The nominal control objective is to regulate the output toward zero.

Source: T. Nauta, H. Sandberg, and M. Maggio, "Stealthy Computational Delay Attacks on Control Systems," ICCPS 2025.

Computational Delay Attack and Switched Dynamics

Augmented closed-loop state

The state includes the plant, controller, and stored information:

$$\psi_k = \begin{bmatrix} x_k \\ \tilde{y}_k \\ z_k \\ u_k \end{bmatrix}$$

$\tilde{y}_k$ is the latest stored measurement. The augmented state ψ_k captures the complete closed-loop state used by the attack model.

Binary deadline decision

$$a_k = \begin{cases} 0, & \text{deadline hit,} \\ 1, & \text{deadline miss.} \end{cases}$$

$$\psi_{k+1} = \left[(1 - a_k)\Phi_h + a_k \Phi_m\right]\psi_k$$

Φ_h and Φ_m are the hit- and miss-mode matrices (paper's switched model, Eq. (7)).

What happens on a missed deadline?

- The physical plant continues evolving.
- Controller computation is delayed; the controller state and stored measurement are not updated.
- The actuator either **HOLDs** the previous command or applies **ZERO**.

Example sequence: 11010

$k=1$	$k=2$	$k=3$	$k=4$	$k=5$
1	1	0	1	0
Miss	Miss	Hit	Miss	Hit

1 = miss; 0 = hit. The attacker chooses timing failures, not the output directly.

Source: T. Nauta, H. Sandberg, and M. Maggio, "Stealthy Computational Delay Attacks on Control Systems," ICCPS 2025.

Optimal Stealthy Attack: Impact Under Constraints

Compact optimization formulation

$$
\begin{aligned}
\max_{a_0,\dots,a_{\ell-1}} \quad & y_\ell^{\mathsf{T}} Q y_\ell \\
\text{subject to} \quad & a_k \in \{0,1\}, \quad k = 0,\dots,\ell-1, \\
& \sum_{q=j}^{j+\bar{a}} a_q \le \bar{a}, \\
& \qquad j = 0,\dots,\ell-\bar{a}-1, \\
& \psi_{k+1} = \big[(1-a_k)\Phi_h + a_k \Phi_m\big]\psi_k, \\
& \text{binomial and geometric stealth constraints.}
\end{aligned}
$$

- a_k: binary attack decisions; ℓ: attack horizon.
- The watchdog constraint limits consecutive misses.
- Gurobi optimizes the feasible miss sequence; it does not choose Q or directly set y_ℓ.

Do not conflate these quantities

$$J_{\text{attack}}(\ell) = y_\ell^{\mathsf{T}} Q y_\ell$$

$$\boxed{\omega(\ell) = J_{\text{attack}}(\ell) - J_{\text{nominal}}(\ell)}$$

Both trajectories start from the same initial conditions. The objective is the quantity maximized; $\omega(\ell)$ measures the additional deviation relative to the no-attack baseline.

Output weighting Q

- Furuta pendulum: $Q = \mathrm{diag}(1,0,0)$ (pendulum angle).
- Quadruple tank: $Q = \mathrm{diag}(1,1)$ (both regulated outputs).

$Q = I$ is a mathematical illustration, not a universal choice.

Source: T. Nauta, H. Sandberg, and M. Maggio, "Stealthy Computational Delay Attacks on Control Systems," ICCPS 2025.

Detection I: Watchdog and Binomial Test

Watchdog: consecutive misses

No more than $\bar{a}$ consecutive misses

- $a_k = 1$ denotes a deadline miss.
- If $\bar{a} = 2$, three consecutive misses violate the watchdog rule.
- Passing the watchdog limits runs, but not necessarily the total number of misses in a window.

Model assumption

Nominal misses are modeled as independent events with estimated probability $\hat{p}_m$. This is an assumption, not a universal property of every real-time system.

Sliding-window binomial test

Let $w_i \in \{0, 1\}$ denote the hit/miss status of job i in a window of n_w jobs, where 1 denotes a miss.

$$S_w = \sum_{i=1}^{n_w} w_i$$

Under the nominal model,

$$S_w \sim \text{Binomial}(n_w, \hat{p}_m).$$

The one-sided hypotheses are

$$H_0 : p \leq \hat{p}_m, \qquad H_1 : p > \hat{p}_m.$$

The sequence passes the binomial test when

$$\boxed{\sum_{i=1}^{n_w} w_i \leq \mathcal{B}_{1-\alpha}(n_w, \hat{p}_m)}$$

$\mathcal{B}_{1-\alpha}$ is the binomial critical count; α is the significance level.

Key distinction: the watchdog limits consecutive misses; the binomial test limits the total miss count in a window. Passing a test does not prove that no attack is present.

Source: T. Nauta, H. Sandberg, and M. Maggio, "Stealthy Computational Delay Attacks on Control Systems," ICCPS 2025.

Detection II: Geometric Run-Length and χ^2 Test

Run-length categories

Let R be the number of consecutive misses before a hit. The final category groups the tail at the watchdog limit:

$$P(R = r) = \begin{cases} \hat{p}_m^{\,r}(1 - \hat{p}_m), & 0 \le r < \bar{a}, \\ \hat{p}_m^{\,\bar{a}}, & r = \bar{a}. \end{cases}$$

The categories are $r = 0, 1, \ldots, \bar{a}$; the final category groups the tail run length.

Expected and observed counts

If N is the total number of completed jobs used in the geometric test, then

$$\boxed{E_r = NP(R = r)}$$

where m_r is the observed count for category r.

Chi-square statistic

$$\boxed{S_{\chi^2} = \sum_{r=0}^{\bar{a}} \frac{(m_r - E_r)^2}{E_r}}$$

The acceptance condition is

$$\boxed{S_{\chi^2} \le \chi^2_{1-\alpha}(\bar{a})}$$

$\chi^2_{1-\alpha}(\bar{a})$ is the critical value at significance level α, using the paper's degrees of freedom.

Why use both statistical tests?

A sequence may satisfy the watchdog yet have an abnormal run-length pattern. The binomial test checks miss counts; the geometric/χ^2 test checks run-length frequencies.

Source: T. Nauta, H. Sandberg, and M. Maggio, "Stealthy Computational Delay Attacks on Control Systems," ICCPS 2025.